

## Homework 10

**Problem 1.** Show that for a  $C^1$  (Frechet) function  $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ , being conformal is equivalent to being complex-differentiable (see definitions in lecture).

**Problem 2.** Show that with  $\gamma_D = \mathbb{R} + iD$  (oriented left to right) where  $D \in \mathbb{R}$

$$\int_{\gamma_D} e^{-z^2}$$

is constant in  $D$ .

**Problem 3.** Let  $\gamma : [0, 1] \rightarrow \mathbb{C} \setminus \{0\}$  be a smooth loop.

1. Show that there exists  $\phi : [0, 1] \rightarrow \mathbb{C}$  such that  $\phi'(t) = \frac{\gamma'(t)}{\gamma(t)}$  for all  $t \in (0, 1)$ . Also show that  $\exp(-\phi)\gamma$  is constant in  $t$ .
2. Show that

$$\frac{1}{i2\pi} \int_0^1 \frac{\gamma'(t)}{\gamma(t)} dt$$

is always an integer. Furthermore, show that if  $\gamma$  can continuously deform into a single point in  $\mathbb{C} \setminus \{0\}$  (i.e.  $\gamma$  does not enclose the origin 0), then this integer is 0.

*Remark.* This integer counts how many times  $\gamma$  winds around 0.

A smooth deformation of  $\gamma$  into a single point is a smooth map  $\Gamma : [0, 1] \times [0, 1] \rightarrow \mathbb{C}$  such that  $\Gamma(t, 0) = \gamma(t) \forall t$  while  $\Gamma(t, 1)$  is constant in  $t$ .